Ex:12 End-End communication At Transport layer

17/9/26

a) Implement echo client server using TCP/UDP Sockets

Aim:-

To implement an echo client - server application using TCP sockets and verify communication between the client and server

steps:-

1. create a TCP Echo server using server and socket
2. Assign a port number to the server.
3. compile and execute the TCP server program
4. create a TCP Echo client using socket
5. connect the client to the server using the server address and port number.
6. Enter a message through the client
7. the server receives the message and sends it back to the client.
8. Display the echoed message at the client
9. Enter bye to terminate the communication

Result:-

Thus, the TCP Echo client - server application was Successfully implemented and the client Successfully received the echoed message from the server.

Program:

TCP Echo server - Java :-

```
import java.io.*;

import java.net.*;

Public class TCP chosrver{

    public static void main(String [] args)

throws Exception{

        ServerSocket SS = new ServerSocket

while ((msg = in.readLine()) != null){

            out.println("Echo"+msg);

if (msg.equals Ignore ("bye")) break;

}

S.close();

SS.close();

}
```

b) Echo client - server using UDP

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Aim:-

To implement an Echo client - server application using UDP Sockets and verify communication between the client and server.

steps:-

1. create a UPD Echo server using Datagram socket and Datagram packet.
2. Assign a port number to the UDP server
3. compile and execute the UDP server program
4. create a UDP Echo client using Data gram Socket and Datagram packet.
5. specify the server address and port number
6. Enter a message through the client.
7. The client sends the message as a UDP datagram to the server.
8. The server receivers the datagram and sends the echoed message back.
9. Display the received echoed message of the client
10. Enter bytie to terminate the communication.

Result:-

Thus, the UPD Echo client - server application was successfully implemented and the client successfully received the echo message from the server.

Program:

UDP Echo server - Java

```
import java.net.*;

Public class UDP Echo server{

Public static void main (string [] args)

        Datagram Socket S = new datagram
                                Socket

while (true) {

Datagram Packet P = new Datagram
S.receive (P);

S.send (new Datagram Packet (r, r.length,
                P.getAddress(), P.get
                port()));

If (msg.equals Ignore case ("bye")) break;}
        S.close(); }

}
```

c) chat client - server using TCP/UDP sockets
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Aim:-

To implement a chat client - server application using sockets and enable communication between multiple clients through a server.

steps:-

1. create a chat server program using socket program
2. start and run the chat server.
3. create the chat client program and connect it to the server.
4. Run two or more client program in seperate termenals.
5. Enter a name of each client
6. send message from one client to the server
7. the server broadcasts the message to the order connected clients.
8. verify that the clients can exchange message through the server.
9. continue the communication between the clients
10. Enter / quit to terminates a client connection.

Result:-

Thus, the chat client - server application was successfully implemented and message was successfully exchanged between multiple clients through the server.

program:

```
import java.io.*;
import java.net.*;
public class chatclient {
public static void main(String[] args)
                    throws exception {
    Socket s=new Socket("127.0.0.1", 5000);
new thread(E) ->{

try {
    string m;
while((m=in.readline())!=null)
system.out.println(m);}
catch(Exception){}

string msg;
while((msg=kb.readline())!=null){
    out.println(msg);
    if(msg.equalsIgnoreCase("/quit")
                    )break;
}
s.close();

}

}
```